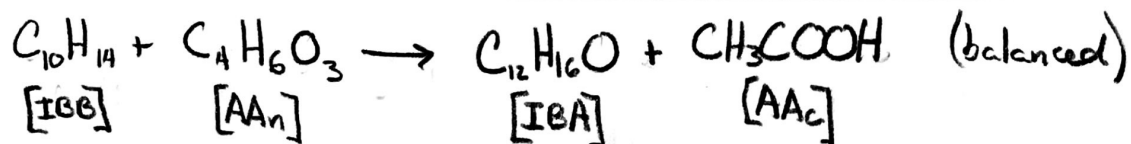


P4.10



Mass in:

$$(134\text{g IBB}) + (134\text{g AAn}) = 268\text{g}$$

Mass out:

$$(27\text{g IBB}) + (52\text{g AAn}) + (121\text{g IBA}) + (68\text{g AAc}) = 268\text{g}$$

Moles in:

$$(134\text{g IBB}) \left(\frac{1\text{mol}}{134\text{g}} \right) = 1\text{mol IBB}$$

$$(134\text{g AAn}) \left(\frac{1\text{mol}}{102\text{g AAn}} \right) = 1.314\text{mol AAn}$$

Moles out:

$$(27\text{g IBB}) \left(\frac{1\text{mol}}{134\text{g}} \right) = 0.20\text{mol IBB}$$

$$\Delta n_{\text{IBB}} = 1.0 - 0.20\text{mol} = 0.80\text{moles}$$

$$n_{\text{AAn, out}} = 1.314\text{mol} - 0.80\text{mol} = 0.514\text{mol}$$

$$\Rightarrow (0.514\text{mol}) \left(\frac{102\text{g}}{\text{mol}} \right) = 52.4\text{g AAn}$$

$$n_{\text{IBA, out}} = 0.80\text{mol} \Rightarrow (0.80\text{mol}) \left(\frac{176\text{g}}{\text{mol}} \right) = 140.8\text{g IBA}$$

$$n_{\text{AAc, out}} = 0.80\text{mol} \Rightarrow (0.80\text{mol}) \left(\frac{60\text{g}}{\text{mol}} \right) = 48\text{g AAc}$$

IBA and AAc don't match!

P.4.19 $\dot{n}_1 = 100 \text{ g/mol}$ (basis)

$$A_c = 0.02$$

$$AA_m = 0.14$$

$$M = 0.36$$

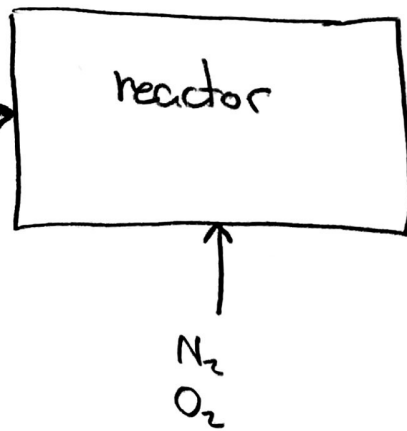
$$E = 0.20$$

$$N_2 = 0.13$$

$$O_2 = 0.04$$

$$CO = 0.08$$

$$H_2S = 0.03$$



C:

$$146 = \dot{n}_{CO_2, out} + \dot{n}_{CO, out}$$

H:

$$183 = \dot{n}_{H_2O, out}$$

S:

$$3 = \dot{n}_{SO_2, out}$$

O:

$$2 \dot{n}_{O_2, in} = 2 \dot{n}_{O_2, out} + \dot{n}_{CO, out} + 2 \dot{n}_{CO_2, out} + 129$$

N:

$$2 \dot{n}_{N_2, in} = 2 \dot{n}_{N_2, out} - 26$$

Assume complete combustion: $\dot{n}_{CO, out} = 0$ and $\dot{n}_{O_2, out} = 0$

$$\dot{n}_{CO_2, out} = 146$$

$$\dot{n}_{O_2, in} = \frac{2(146) + 129}{2} = 210.5$$

using ratio composition of air $\frac{\dot{n}_{O_2, in}}{\dot{n}_{N_2, in}} = \frac{21}{79}$

$$\dot{n}_{N_2, in} = \frac{79}{21} (210.5) = 792$$

Now for actual case: (20% excess air)

$$\dot{n}_{O_2, in} = 1.2(210.5) = 252.6 \frac{\text{gmol}}{\text{min}}$$

$$\dot{n}_{N_2, in} = 1.2(792) = 950.4 \frac{\text{gmol}}{\text{min}}$$

know ratio of $\frac{\dot{n}_{CO_2, out}}{\dot{n}_{CO, out}} = \frac{4}{1} \Rightarrow \frac{\dot{n}_{CO_2, out}}{\dot{n}_{CO, out}} = \frac{4(28)}{44} = \frac{2.55 \text{ mol CO}_2}{\text{mol CO}}$

N:

$$\dot{n}_{N_2, out} = \frac{2(950.4) - 26}{2} = 963 \frac{\text{gmol}}{\text{min}}$$

C:

$$146 = 2.55 \dot{n}_{CO, out} + \dot{n}_{CO, out} \Rightarrow \dot{n}_{CO, out} = \frac{146}{3.55} = 41.13 \frac{\text{gmol}}{\text{min}}$$

$$146 = \dot{n}_{CO_2, out} + 41.13 \Rightarrow \dot{n}_{CO_2, out} = 104.87 \frac{\text{gmol}}{\text{min}}$$

O:

$$\dot{n}_{O_2, out} = \frac{2(252.6) - 41.13 - 2(104.87) - 129}{2}$$

$$\Rightarrow \dot{n}_{O_2, out} = 62.67 \frac{\text{gmol}}{\text{min}}$$

Total moles: